

Assignment Problem Statement: AI-Powered Product Strategy Assistant

Overview

Product Managers are responsible for making critical business decisions that shape the future of a product. To make informed decisions, they must analyze information from multiple sources such as customer feedback, feature requests, product reviews, sales reports, product usage analytics, competitor research, and market trends.

As organizations grow, the volume of available information increases significantly. Product teams often spend considerable time manually reviewing documents, collecting insights, comparing competitors, prioritizing features, and preparing strategy reports. This process is often slow, repetitive, difficult to scale, and may lead to inconsistent decision-making.

The organization wants to leverage Generative AI to build an intelligent Product Strategy Assistant capable of analyzing business information, generating actionable insights, and assisting Product Managers in making data-driven strategic decisions.

Objective

Design and develop an AI-powered Product Strategy Assistant that helps Product Managers analyze business information, identify opportunities, prioritize initiatives, and generate strategic recommendations.

The solution should utilize multiple AI agents working together to perform various business analysis tasks and deliver meaningful insights through an interactive application.

Expected Solution

The proposed system should be able to:

- Understand product-related questions in natural language.
 - Process structured and unstructured business information.
 - Analyze customer feedback and feature requests.
 - Evaluate market and competitor information.
 - Generate strategic product insights.
 - Assist with feature prioritization.
 - Support roadmap planning activities.
 - Produce executive-level summaries and reports.
 - Provide an interactive chat-based experience for users.
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Overall Solution Flow

Step 1: Data Ingestion

Users upload one or more data sources such as:

- Customer reviews
- Survey responses
- Feature requests
- Sales reports
- Product analytics
- Market research documents
- Competitor information

Step 2: Multi-Agent Processing

The uploaded information is processed by multiple AI agents.

Step 3: Agent Collaboration

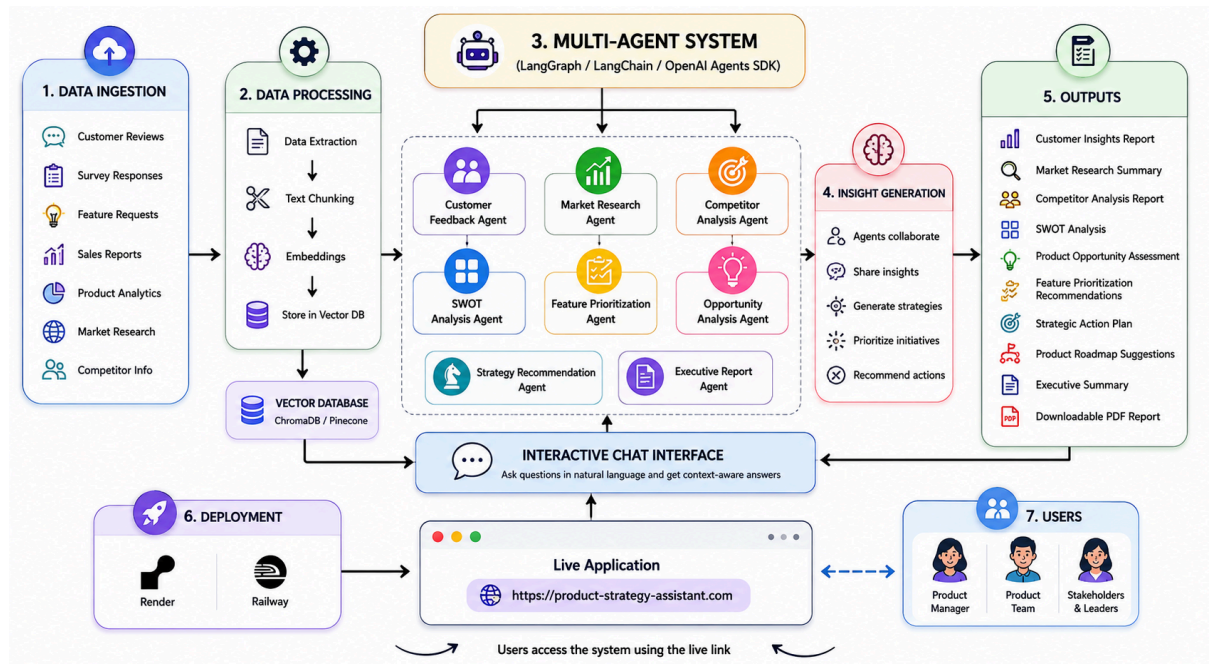
Agents analyze different aspects of the data and share insights with one another.

Step 4: Insight Generation

The system generates strategic business insights and recommendations.

Step 5: Report Creation

The final outputs are consolidated into a downloadable executive report.



AI Agent Requirements

Participants are expected to implement a Multi-Agent architecture.

Guidelines

- Minimum Agents: 3
- Recommended Agents: 5–7
- Maximum Agents: No Limit

Participants are free to decide:

- Number of agents
- Agent responsibilities
- Agent communication mechanisms

- Workflow orchestration design
- Analysis methodologies
- Report generation approach

Example agent roles may include:

- Customer Feedback Agent
- Market Research Agent
- Competitor Analysis Agent
- SWOT Analysis Agent
- Feature Prioritization Agent
- Opportunity Analysis Agent
- Strategy Recommendation Agent
- Executive Report Agent

These are suggestions only. Participants are encouraged to design their own agent architecture.

Expected Outputs

The application should generate some or all of the following outputs:

- Customer Insights Report
- Market Research Summary
- Competitor Analysis Report
- SWOT Analysis
- Product Opportunity Assessment
- Feature Prioritization Recommendations
- Strategic Action Plan
- Product Roadmap Suggestions
- Executive Summary
- Downloadable PDF Report

Additionally, users should be able to interact with the system through natural language queries and receive context-aware responses.

Suggested Technologies

Participants may use:

AI Models

- GPT-4o Mini
- text-embedding-3-small

Frameworks

- LangGraph
- LangChain

Vector Database

- ChromaDB / Pinecone

Backend

- FastAPI

Frontend

- Streamlit / React / HTML

Deployment

- Render or Railway

Alternative technologies may also be used.

Evaluation Criteria

1. Successful Deployment (30%)

2. Quality of AI Insights (35%)

3. Multi-Agent Design and User Experience (35%)

Submission Requirements

Participants must submit:

1. Source Code Repository
 2. Live Application URL
 3. Architecture Diagram
 4. Sample Generated Report (PDF)
 5. Project Documentation
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Bonus Features

Additional consideration may be given for implementing:

- Advanced Multi-Agent Collaboration
 - Product Opportunity Scoring
 - Roadmap Generation
 - Interactive Dashboards
 - Executive Presentation Generation
 - Evaluation and Monitoring Frameworks
 - Voice-Based Interaction
 - Innovative Product Strategy Methodologies
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Expected Outcome

The final solution should function as an AI-powered Product Strategy Assistant capable of helping Product Managers make faster, smarter, and more data-driven decisions by transforming large volumes of business information into actionable strategic insights.